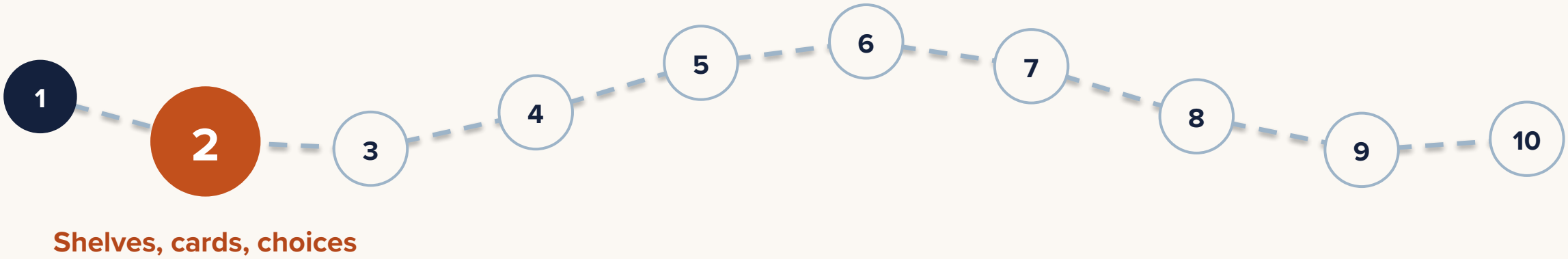
PYTHON FOR SUPPLY CHAIN

Session 2 · Wed 18 Nov 2026

Shelves, cards, choices

lists, dictionaries, comparisons, if rules



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Today

1

Lists

`[6, 6, 8]`

2

Count and total

`len` `sum` `min` `max`

3

Dictionaries

`{"quantity": 6}`

4

Choices

`==` `if` `elif` `else`

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A list is a row of bins

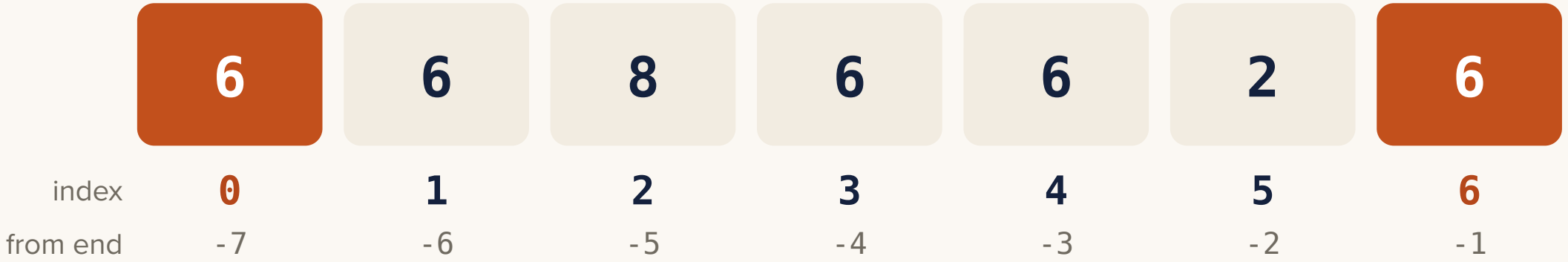



```
quantities = [6, 6, 8, 6, 6, 2, 6]
```

Like one column in Excel

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Python counts from 0




```
quantities[0]
→ 6
```

```
quantities[-1]
→ 6
```

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Count, total, smallest, largest

`len(quantities)`

7

Excel: COUNT

`sum(quantities)`

40

Excel: SUM

`min(quantities)`

2

Excel: MIN

`max(quantities)`

8

Excel: MAX

Same idea, new spelling.

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A dictionary is a product card

key	value
`stock_code`	`"85123A"`
`quantity`	`6`
`price`	`2.55`
`country`	`"United Kingdom"`

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Look up by name, not by position

```
order_line["quantity"]
```

```
→ 6
```

```
order_line["Quantity"]
```

```
→ KeyError: 'Quantity'
```

The key must match exactly.

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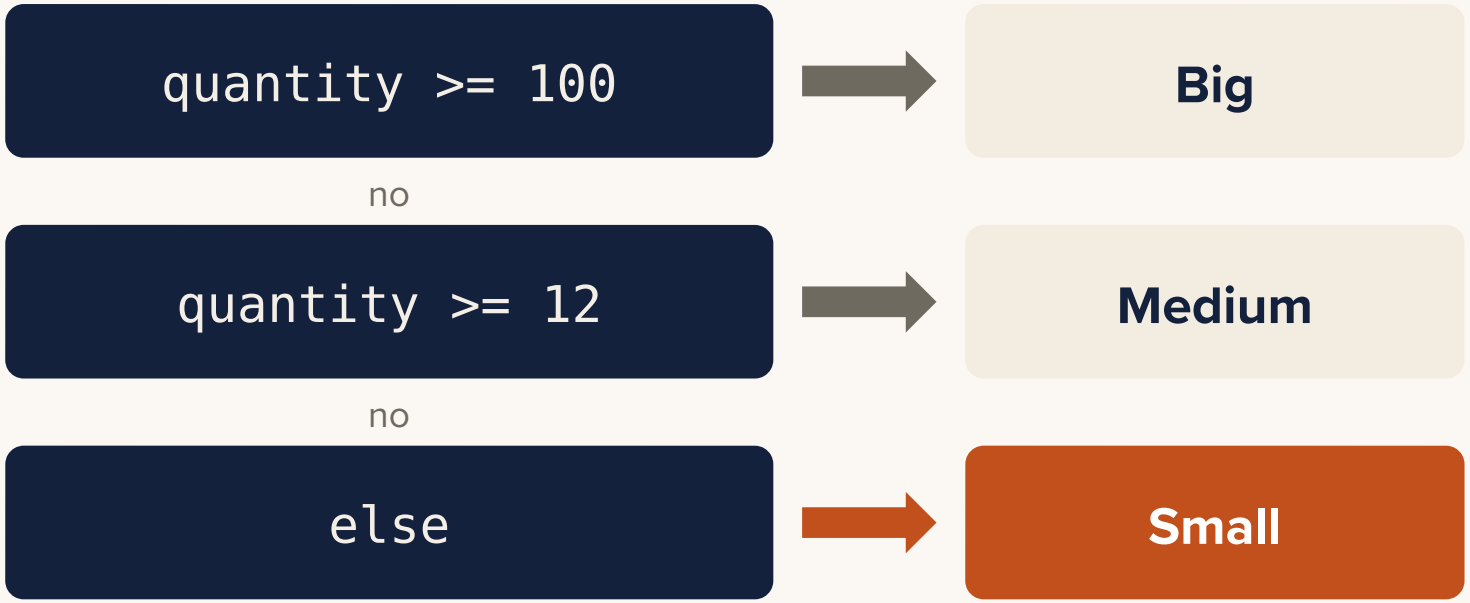
A comparison answers True or False



=	==
stores	asks

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if, elif, else: a packing rule



quantity = 6: only one branch runs.

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The colon and the indent

```
if quantity >= 100:
    print("Big order line")
elif quantity >= 12:
    print("Medium order line")
else:
    print("Small order line")
```

: ends it

4 spaces

No colon or no indent: the rule breaks.

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A big line, not a big order

Invoice 536365		
WHITE HANGING HEART T-LIGHT HOLDER	6	
WHITE METAL LANTERN	6	
CREAM CUPID HEARTS COAT HANGER	8	

one row = one product on one invoice

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Three errors you will meet

`IndexError`

no bin at that position

```
quantities[7]
```

`KeyError`

no field with that name

```
order_line["Price"]
```

`SyntaxError`

the colon is missing

```
if quantity > 12
```

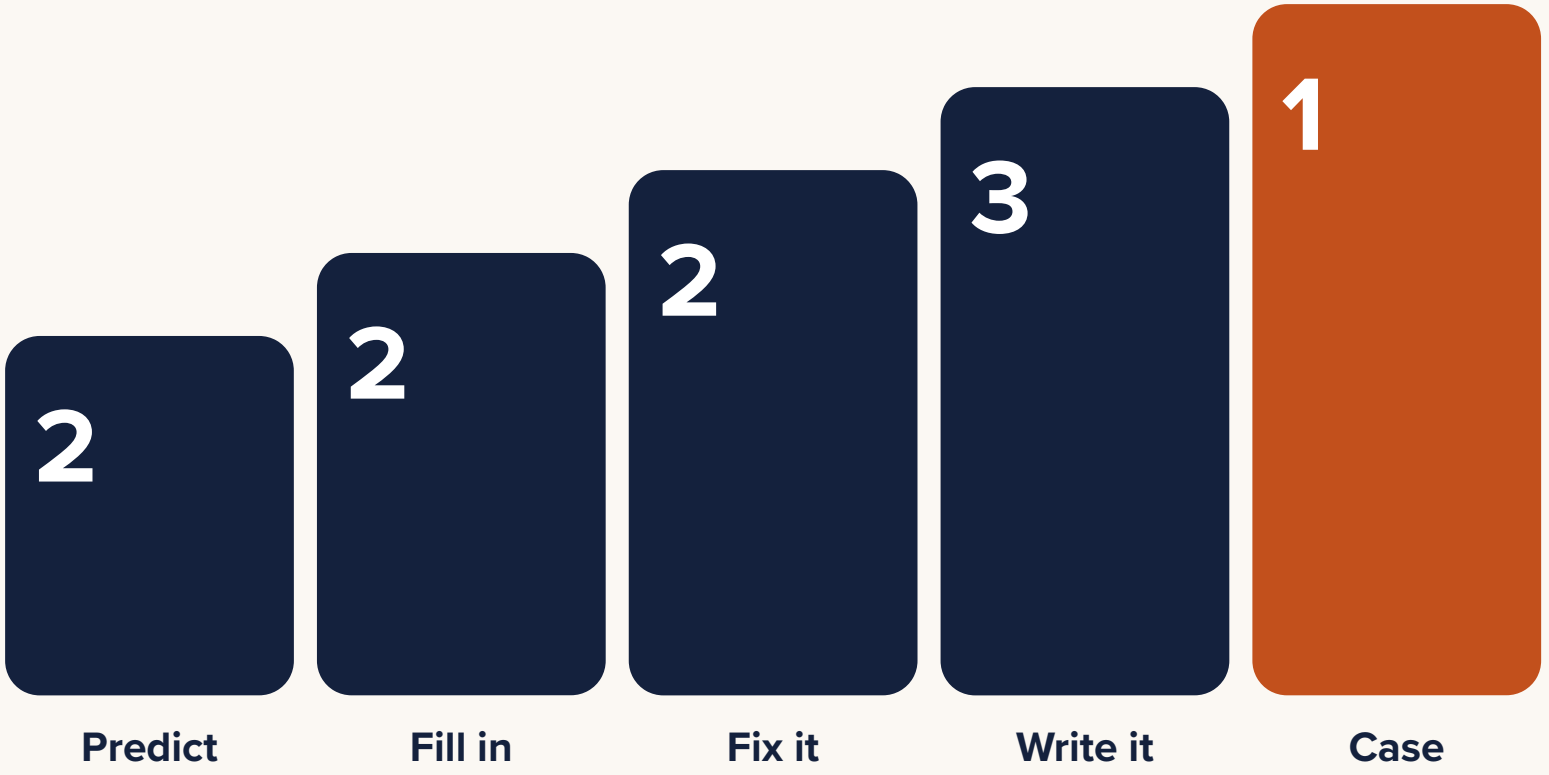
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Your toolkit so far

Code	Does
`[6, 6, 8]`	a list: values in order
`quantities[0]`	the first value
`quantities[-1]`	the last value
`len  sum  min  max`	count, total, smallest, largest
`{"quantity": 6}`	a dictionary: values by name
`order_line["quantity"]`	look up by name
`==  !=  >  >=`	compare, gives True or False
`if / elif / else`	choose one branch

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Now in Colab



Notebook

s2_shelves_cards

Homework

2 extras

Recall

Thu 19 to Sun 22

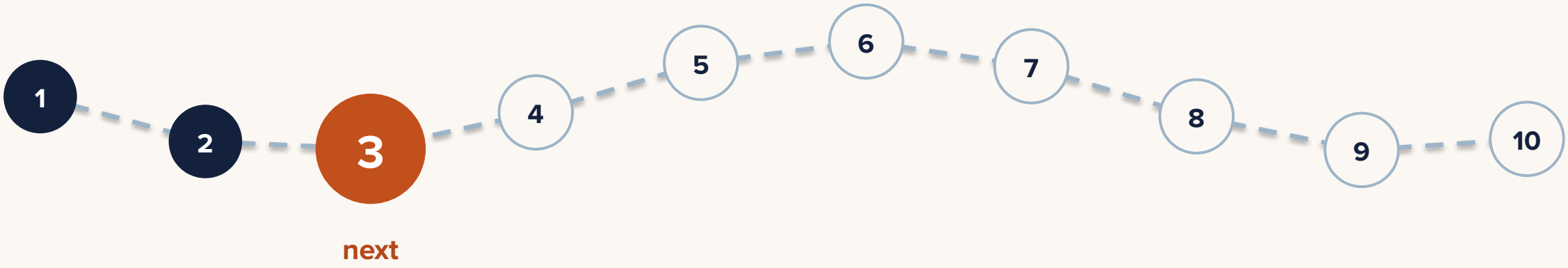
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Thank you

Next: Session 3

Can one rule label every line?

Mon 23 Nov



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